PREDICTING DEFAULT RISK ON PEER-TO-PEER LENDING PLATFORM

BUSA 310A Business Analytics III: Predictive and Prescriptive Business Analytics

Professor Yanchao Yang | Spring 2025 | April 13, 2025

Vy (Valerie) Le

Abstract

This study investigates the factors that predict default risk in peer-to-peer (P2P) lending, a rapidly growing financial model that bypasses traditional banking institutions. Using a comprehensive dataset of over 600,000 loan records, we examine how borrower characteristics—such as income level, debt-to-income ratio, FICO score, loan purpose, home ownership, and loan term—affect the likelihood of loan default. We employ a range of regression-based machine learning models, including Ordinary Least Squares (OLS), Logistic Regression, Ridge, and Lasso, to classify loans as either defaulted or fully paid. Our analysis begins with detailed exploratory data visualizations to uncover key risk patterns, followed by model evaluations using accuracy rates and confusion matrices. Among the methods tested, Lasso Regression yields the highest predictive accuracy at 79.35%, highlighting the value of regularization in high-dimensional financial data. These results not only provide insight into the structure of credit risk in P2P markets but also offer practical implications for investors seeking to optimize lending decisions. This project demonstrates how data-driven approaches can support more efficient, fair, and informed financial ecosystems.

Keywords: *Default Risk Prediction, Peer-to-Peer Lending, Logistic and Regularized Regression, Credit Risk Analytics, Data-Driven Investment Strategy*

1. Introduction

In recent years, peer-to-peer (P2P) lending has emerged as a disruptive force in the financial services sector, offering an alternative channel for individuals to access credit without traditional

banking intermediaries. By leveraging online platforms, P2P lending democratizes both borrowing and investing—enabling borrowers with diverse financial backgrounds to secure loans, and allowing investors to earn returns by funding these loans directly. This disintermediation fosters financial inclusion and greater autonomy in credit markets. However, it also introduces significant uncertainty. Without the institutional safeguards of banks—such as centralized underwriting, regulation, and capital buffers—the risk of borrower default becomes a central concern for platform sustainability and investor confidence.

Unlike traditional lending, which often relies on deep financial histories and institutional relationships, P2P lending decisions are made based on limited, often self-reported, data. This makes the ability to accurately assess creditworthiness a cornerstone of effective platform design. If investors cannot differentiate between high-risk and low-risk loans, market inefficiencies emerge, resulting in capital misallocation, higher default rates, and declining trust in the P2P model.

Understanding and predicting default risk is therefore essential for ensuring both the financial health of P2P platforms and the success of individual investment strategies. This project seeks to answer a key question: "*Can we accurately predict loan default risk using historical data from a P2P lending platform?"* To address this, we analyze a high-dimensional dataset of past loan applications, borrower characteristics, and repayment outcomes. Our objective is to build robust predictive models that can classify whether a loan will default or be fully repaid—enabling investors to make more informed, data-driven decisions.

Through a combination of exploratory data analysis (EDA) and machine learning models—including Ordinary Least Squares (OLS), Logistic Regression, Ridge Regression, and Lasso Regression—we evaluate which borrower attributes (e.g., income, debt-to-income ratio, FICO score, loan purpose, and loan term) are most predictive of default. We assess each model's performance using confusion matrices and accuracy metrics, with a particular focus on identifying models that balance predictive power, interpretability, and generalizability.

The importance of accurate default prediction in P2P lending has been explored in several recent studies, each highlighting different aspects of model performance, feature importance, and data structure. For instance, Serrano-Cinca et al. (2015) conducted one of the earliest large-scale empirical studies using Lending Club data, showing that factors such as loan purpose, annual income, housing status, credit history, and borrower indebtedness were significant predictors of

default. Their use of logistic regression and survival analysis confirmed that borrower characteristics could meaningfully improve default prediction beyond the platform-assigned loan grades.

Similarly, Chang et al. (2022) built machine learning models on Lending Club data using algorithms such as logistic regression, decision trees, support vector machines, random forests, XGBoost, LightGBM, and neural networks. Their results demonstrated that ensemble learning methods—particularly XGBoost and LightGBM—outperformed traditional credit scoring techniques in both predictive accuracy and efficiency. They also emphasized the role of feature engineering in improving model quality and interpretability.

Expanding beyond P2P platforms, Munkhdalai et al. (2019) compared expert-based scoring systems like FICO with machine learning models using real U.S. consumer data. They applied algorithms including logistic regression, support vector machines, XGBoost, and deep neural networks, and concluded that machine learning methods significantly outperformed traditional scoring models when paired with robust feature selection strategies. Their use of Random Forest-based selection techniques (TSFFS and NAP) highlighted the value of filtering high-dimensional data for optimal model performance.

These studies collectively support our approach: using interpretable yet powerful regression models to predict loan defaults based on borrower-level features. They reinforce the need for robust model selection, variable preprocessing, and regularization techniques when working with complex real-world lending data.

Building on this foundation, our study contributes by systematically comparing traditional regression methods with regularized models in a controlled, interpretable framework. While prior work has explored black-box models with high performance, our focus includes both predictive accuracy and transparency—key considerations for real-world decision-makers. Ultimately, this analysis contributes to a deeper understanding of credit risk in P2P lending markets, revealing both opportunities for optimized loan allocation and challenges in forecasting borrower behavior. By combining analytical rigor with business insight, we aim to provide not just technical predictions, but actionable recommendations for investors and platform designers alike.

2. Data Section

Before any modeling or analysis could take place, our first step was to clean, explore, and prepare the dataset. The data comes from a peer-to-peer (P2P) lending platform and contains hundreds of thousands of individual loan records, each accompanied by borrower and loan-level features such as credit score, debt-to-income ratio, income, loan term, loan purpose, and repayment outcome. The variable of greatest importance at this stage was loan_status, which reports the final or current condition of each loan—whether it was paid back in full, charged off as bad debt, defaulted, or still in progress (e.g., current, in grace period, or late).

To begin understanding this variable, we visualized the full distribution of loan statuses across the dataset. As shown in Figure 1, the majority of loans fell into the "Fully Paid" or "Current" categories. A significant portion, however, were labeled as "Charged Off," indicating that the borrower stopped repaying and the platform no longer expected to collect on the loan. A smaller number were marked as "Late" or "In Grace Period," and just a handful were recorded as formally "Defaulted." While useful for understanding borrower performance at a glance, many of these categories did not represent completed repayment outcomes. Since our goal was to build predictive models for default risk, we needed to narrow the dataset to finalized loan outcomes only.

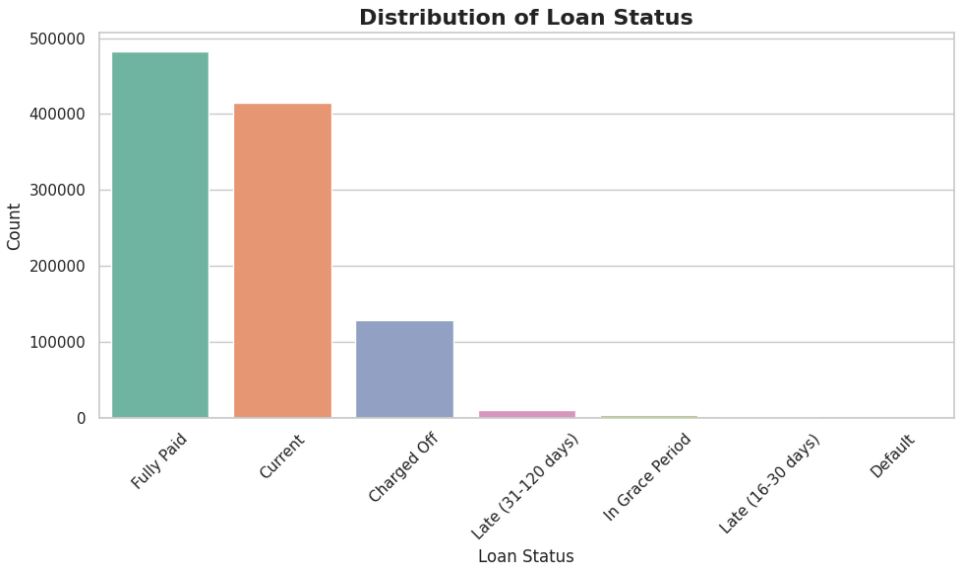

Figure 1: Distribution of Loan Status (Raw)

Therefore, we filtered the dataset to retain only those loans that had one of three outcomes: "Fully Paid," "Charged Off," or "Default." All other statuses were excluded, as they represented loans still in progress or with unresolved payment behavior. This decision ensured that our

analysis would be based solely on completed loans, where the true repayment outcome was known. The updated distribution after filtering is visualized in Figure 2. As we can see, the remaining dataset consists primarily of fully paid loans, followed by a smaller (but still significant) number of charged-off loans. While loans marked as "Default" were technically present, they were extremely rare compared to the other two categories.

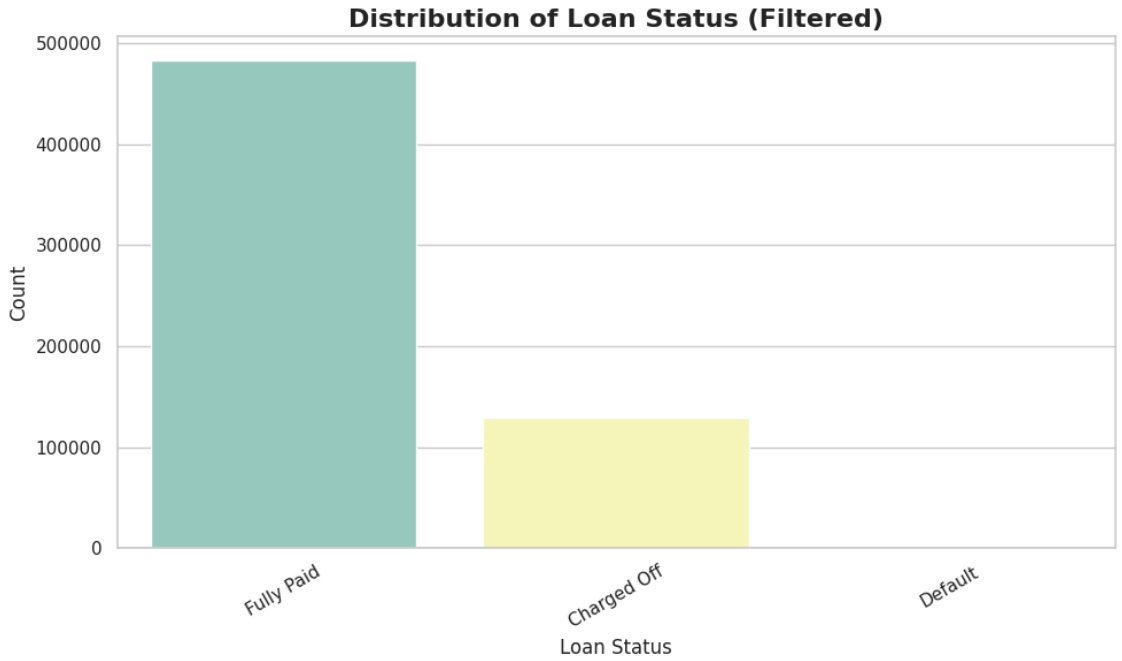

Figure 2: Distribution of Loan Status (Filtered)

Next, we needed to convert the filtered loan_status variable into a binary format that could serve as the target variable for classification modeling. We created a new dummy variable, loan_default, to indicate whether or not a loan ended in default. Loans marked as "Charged Off" or "Default" were assigned a value of 1 (indicating default), and those labeled as "Fully Paid" were assigned a 0 (indicating successful repayment). This binary outcome variable simplified the modeling process and made it possible to train classification models that distinguish between good and bad loans. Figure 3 shows the distribution of this new variable: roughly 21% of the loans in our working dataset ended in default (charged-off), while nearly 79% were fully repaid. This means that about one in every five loans failed to meet repayment obligations—a substantial rate that underscores the importance of effective risk assessment in the P2P lending market.

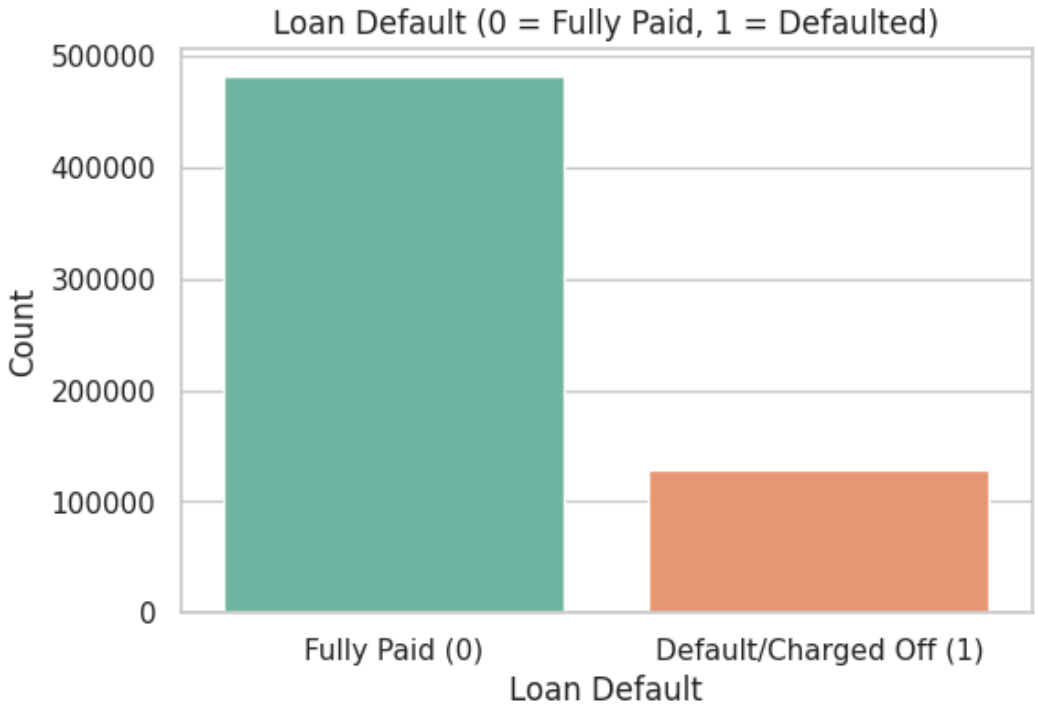

Figure 3: Loan Default Indicator (0 = Fully Paid, 1 = Defaulted)

To better understand the structure of the dataset, we also examined the time period covered by the loans. The dataset spans four years, from 2015 to 2018, providing a snapshot of lending activity and borrower performance over time. We grouped the data by year to determine how many loans were issued each year and how the default rate varied. As visualized in Figure 4, the highest volume of loans was issued in 2015, with issuance dropping sharply in 2016 and continuing to decline through 2018. At the same time, the default rate followed the opposite trend: it rose steadily from 2015 to a peak in 2017 (around 24%) before dropping in 2018 to about 16%.

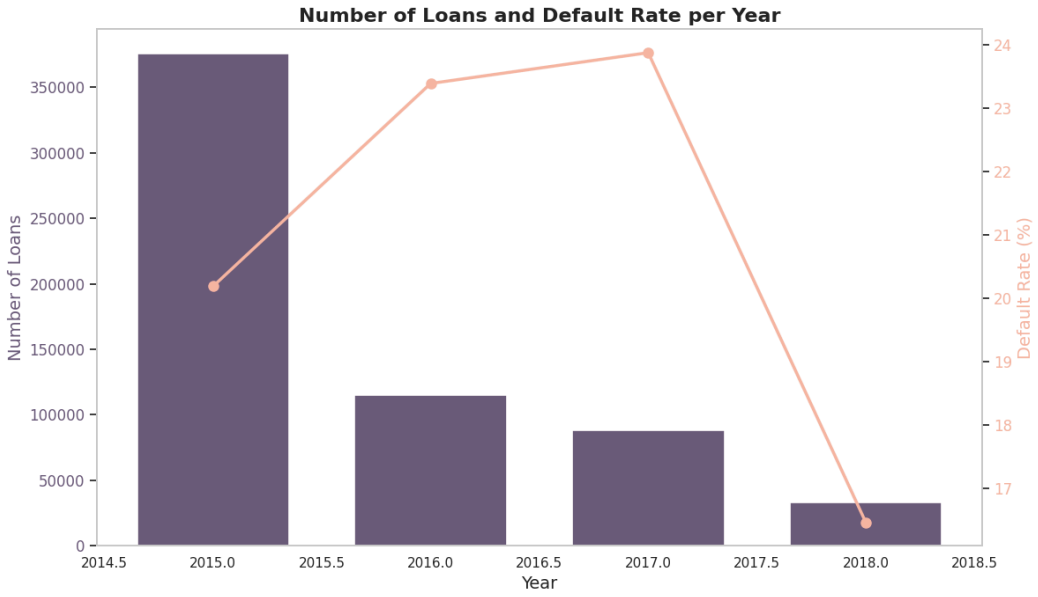

Figure 4: Number of Loans and Default Rate by Year

Year	Number of Loans	Default Rate (%)
2015	375,546	20.19%
2016	115,072	23.39%
2017	88,346	23.87%
2018	32,839	16.46%

Figure 5: Number of Loans and Default Rate by Year (Summary Table)

As shown in the table above, the number of loans issued declined each year from 2015 to 2018. Meanwhile, the default rate increased during the first three years—peaking in 2017 at nearly 24%—before dropping in 2018 to around 16%. This pattern reflects a shift in lending behavior or borrower quality over time, and it highlights the importance of factoring in issuance year when assessing default risk.

This inverse relationship between loan volume and default rate is worth noting. It suggests that the platform may have tightened its credit screening criteria in later years, or that broader economic conditions led to more conservative borrowing and lending behavior. It could also reflect a growing awareness among investors of the need to avoid high-risk loans, particularly

after experiencing higher-than-expected default rates in earlier periods. Regardless of the cause, this trend illustrates the value of conducting time-based analyses when modeling credit risk.

In summary, our data pre-processing workflow included exploring the raw loan status distribution, filtering the dataset to focus on completed loans, creating a binary target variable (loan_default), and analyzing loan issuance trends over time. These steps provided a clean and focused dataset of over 600,000 resolved loans with a clearly defined outcome. With an overall default rate of 21.12% and variation in performance across the years, this data foundation enables us to proceed confidently into deeper exploratory analysis and predictive modeling in the sections that follow.

3. Method Section

In order to conduct our group research, we first analyzed the given dataset and made note of any extraneous values or familiar variables. The next step was to import the dataset into Google Colab and start the data cleaning process. To clean the data, we first noted all values under the variables 'Charged-Off' , 'Default' , and 'Fully Paid.' All other values / variables were deleted. Next, a dummy variable was created for 'Loan Status.' The dummy = 1 if loan_status is charged off. And dummy = 0 if loan_status is fully paid. The overall Charge-Off rate (i.e. default rate) was observed to be 21.12%. This is a fairly high default rate, and can pose as a potential risk for lenders (lenders typically look for default rates under 10%; anything above 10% can imply high risk pool or borrowers).

In this dataset, 4 years are examined - 2015, 2016, 2017, 2018. The number of loans issued per year started out at 375,546 in 2015 and then decreased each year until 2018 when 32,839 loans were issued. The number of loans issued each year significantly decreased year-after-year from 2015 to 2018. On the other hand, the default rate has proved to be much more variable. In 2015, the default rate was 20.19%. Then in 2016, it increased to 23.39% and 23.87% in 2017. Finally, in 2018, it decreased to 16.46%. Using this information, it can be concluded that loans issued in 2016 and 2017 are significantly riskier than loans issued in 2015 and 2018.

Next, to begin the EDA we created a new variable - Income_cat. This variable is split into three categories - Low Income (> 20,000), Medium Income (20,001 - 80,000), and High Income (> 80,000). The next variable created was DTI_cat. This variable assesses an individual's Debt-to-Income (DTI) ratio. This ratio is a measure that compares an individual's monthly debt

payments to their monthly gross income. It's a key indicator that lenders use to evaluate an applicant's ability to manage their monthly payments and repay borrowers' money. DTI_cat is split into three categories - Good (< 28%), Fair (28% - 40%), and Bad (> 40%). The next thing we did was create another variable - FICO_cat. This variable assesses the individual's FICO score and rates them by four different categories - Poor (<580), Fair (580 - 669), Good (670 - 739), and Very good (> 740).

Once all data was collected / cleaned / sorted, and all variables were created, we were prepared to create and run our regression models. In total, we created and ran 4 different regression models - OLS Regression, Logistic Regression, Ridge Regression, and Lasso Regression. In preparation for running the OLS Regression, we first deleted all variables with too many missing values to ensure our model ran as smoothly and effectively as possible. In addition to data cleaning, we also created a dummy variable.

For the Logistic Regression, we used all of the same variables as the OLS Regression previously. We also predicted the probabilities of default status - if probabilities were > 50%, classify as default otherwise, classify as non-default. Additionally, we created a confusion matrix dealing with default status and calculated the prediction accuracy rate of our model by splitting the sample into a test set and a training set and completing a validation test.

Next, to run the Ridge Regression, we used the same variables but used multiple values - 0.001, 0.01, 0.1, 1, 10, 100, 1,000, 10,000. Then, we evaluated the prediction accuracy of the model with similar techniques used previously.

Finally, to run the Lasso Regression, we used the same variables as before and also used multiple values - 0.001, 0.01, 0.1, 1, 10, 100, 1,000, 10,000. The same methods were used to calculate the prediction accuracy of the Lasso Regression model.

4. Results

This section provides a detailed examination of the key findings derived from both the exploratory data analysis and the implementation of predictive models on the peer-to-peer lending dataset.

The primary objective is to assess how effectively various machine learning methods—Ordinary Least Squares (OLS) regression, Logistic regression, Ridge regression, and Lasso regression—can predict loan default. The goal is to identify which model performs best in terms of accuracy, interpretability, and generalizability across a high-dimensional financial dataset.

The analysis begins with a review of key visualizations that reveal important relationships between borrower characteristics and default behavior. These include charts examining loan grade, purpose, term, and home ownership status, providing intuitive insights into patterns that influence loan performance. Following the visual analysis, each predictive model is evaluated based on its classification accuracy, confusion matrix, and feature significance. This structure integrates both visual and quantitative perspectives to offer a comprehensive understanding of the dataset and model performance.

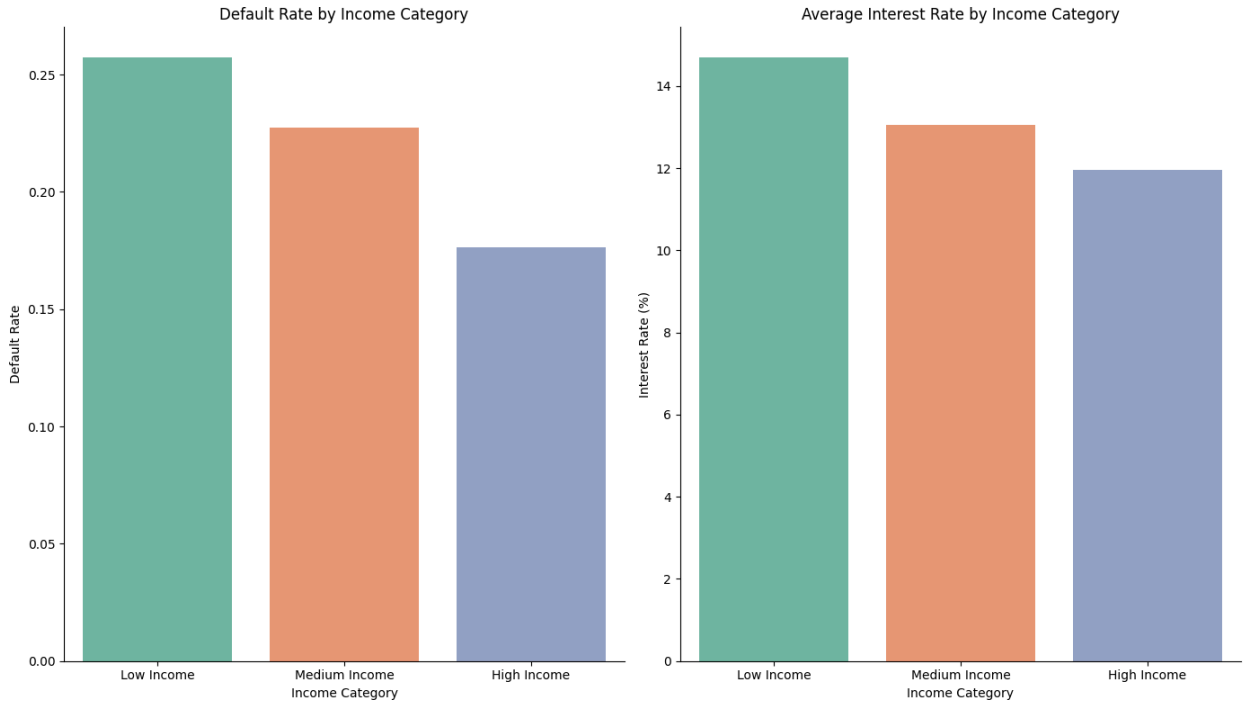

Figure 6

The first set of charts illustrates the relationship between borrower income category and two critical loan performance indicators: default rate and average interest rate. Borrowers were segmented into three income categories—Low Income, Medium Income, and High Income—for comparison.

The left chart shows that default rates are inversely related to income level. Borrowers in the Low Income category exhibit the highest default rate, approximately 26%, followed by Medium Income borrowers at around 23%. High Income borrowers show the lowest default rate, near 18%. This trend is consistent with lending risk profiles, as lower-income individuals typically face higher financial stress and limited repayment capacity, increasing their likelihood of default.

The right chart demonstrates a parallel pattern in interest rate allocation. Low Income borrowers are charged the highest average interest rates, followed by Medium and then High Income groups. This aligns with standard risk-based pricing practices, where lenders assign higher interest rates to compensate for higher perceived credit risk.

Together, these charts confirm that income level is a significant determinant of both loan pricing and repayment behavior. The alignment between default probability and interest rates suggests that the lending platform incorporates income-related risk when setting rates, though the gap in default rates implies there may still be room for improved risk differentiation.

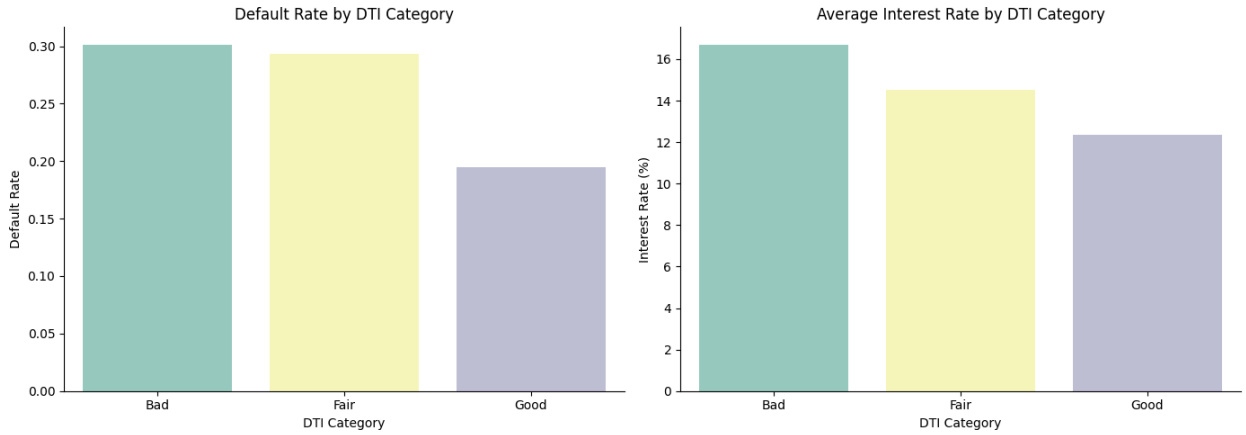

Figure 7

This figure examines the association between borrowers' Debt-to-Income (DTI) ratios and two key lending outcomes: default rate and average interest rate. Borrowers are classified into three categories based on their DTI level: Good, Fair, and Bad.

The default rate is observed to increase as DTI worsens. Borrowers in the Bad DTI category exhibit the highest likelihood of default, with a rate exceeding 30%, followed closely by the Fair group. In contrast, the Good category—representing borrowers with relatively low debt

obligations—shows a substantially lower default rate of approximately 19%. This suggests that DTI is a strong predictor of a borrower's capacity to manage loan repayments.

A corresponding trend is found in interest rate allocation. Borrowers with poor DTI scores are subject to the highest average interest rates (above 16%), whereas those in the Good category benefit from lower rates (around 12%). This reflects a risk-based pricing strategy, where interest rates increase to compensate for heightened credit risk.

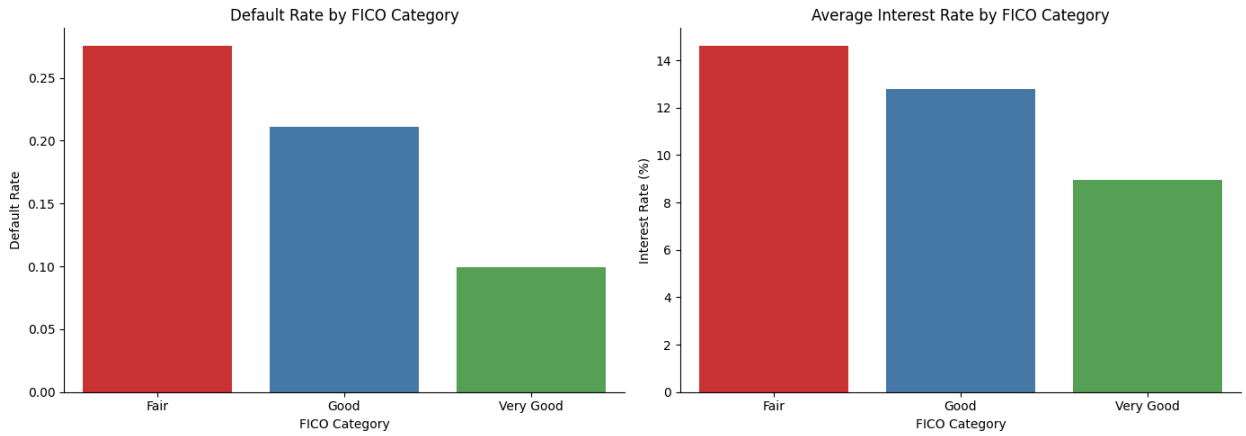

Figure 8

This visualization presents the relationship between borrowers' FICO credit score categories—Fair, Good, and Very Good—and two key financial metrics: default rate and average interest rate.

As expected, borrowers with higher FICO scores tend to exhibit lower default rates. The Fair category shows the highest default rate at approximately 27%, followed by the Good group at around 21%. In contrast, the Very Good category demonstrates significantly lower default activity, with a rate near 10%. This clearly indicates that credit score is a reliable indicator of repayment behavior.

Interest rates also reflect this trend. Borrowers in the Fair category face the highest average interest rates (around 14.6%), while those with Very Good credit are charged the lowest (just under 9%). This suggests that lenders are effectively incorporating creditworthiness into pricing models, offering more favorable terms to less risky borrowers.

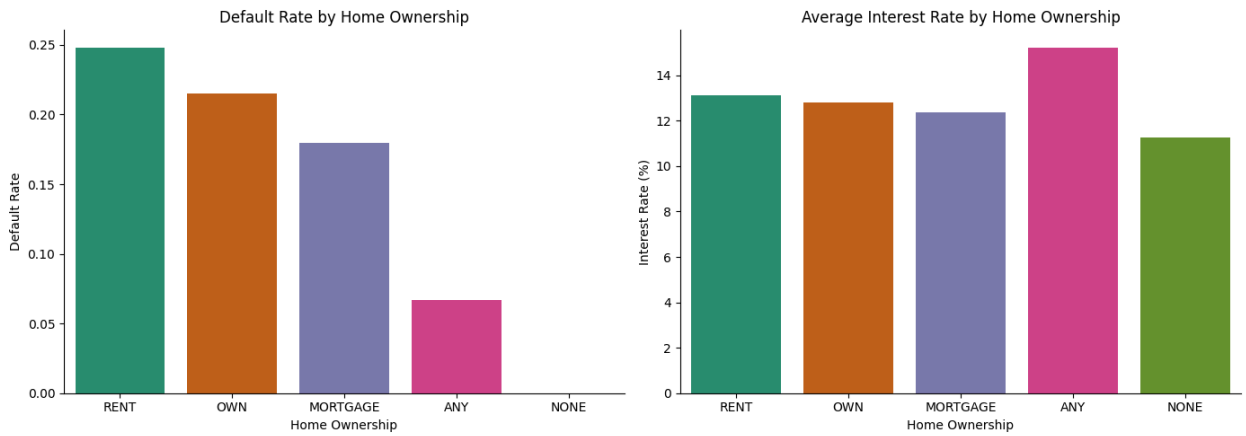

Figure 9

This figure compares default and interest rates across different home ownership statuses. Borrowers who rent exhibit the highest default rate (approximately 25%), followed by those who own (21%) and those with mortgages (18%). Individuals categorized under ANY have a notably lower default rate, though this group is likely underrepresented or irregularly defined.

Interest rates align with the observed risk pattern. Renters face the highest average rates, while mortgage holders benefit from more favorable terms. The ANY group, despite low default rates, receives the highest rates, suggesting potential data irregularities or limited underwriting confidence.

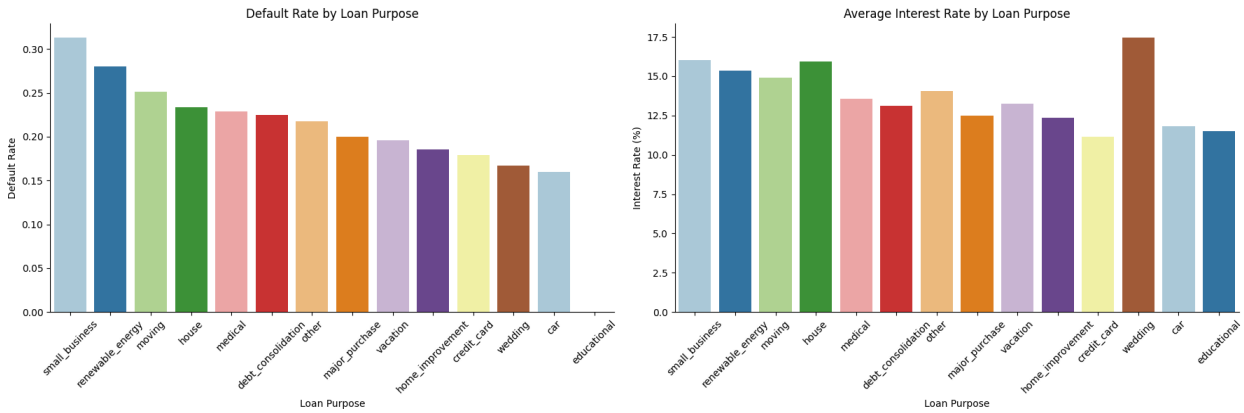

Figure 10

This visualization explores how loan purpose correlates with default risk and pricing. Among all categories, loans for small business and renewable energy purposes display the highest default

rates, both exceeding 27%, followed by moving and house-related purposes. In contrast, educational, car, and wedding loans exhibit lower default rates, generally below 18%.

Interest rate patterns follow a partially aligned structure. Small business and house loans receive the highest average rates, while car and educational loans are priced more conservatively. Notably, wedding loans carry the highest interest rate despite having one of the lowest default rates, possibly reflecting a lack of collateral or limited lender confidence.

These findings suggest that loan purpose serves as a nuanced but informative indicator of credit risk. Certain purposes, particularly small business and renewable energy, are riskier and priced accordingly, while others may be misaligned, pointing to possible inefficiencies or data limitations in pricing models.

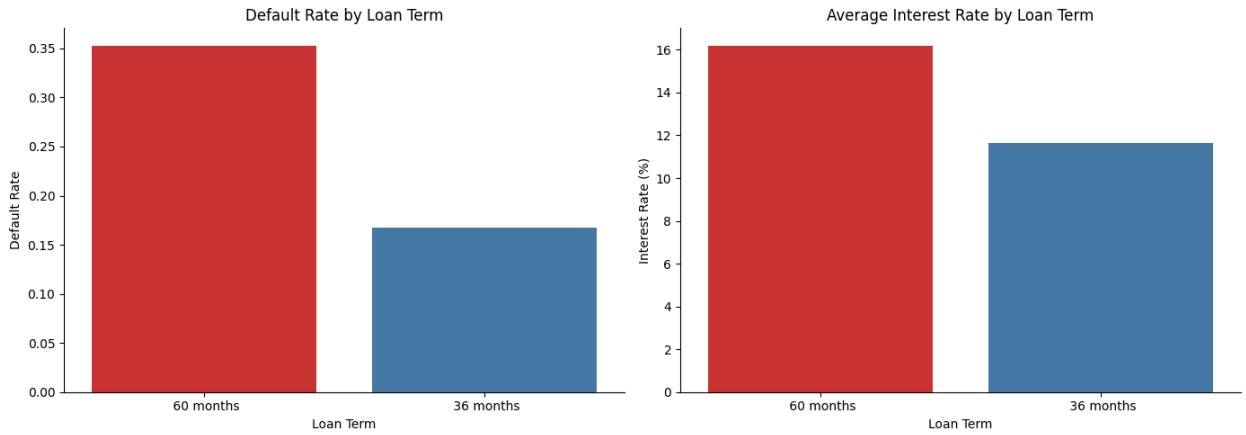

Figure 11

This chart illustrates the impact of loan term length—either 36 or 60 months—on both default probability and average interest rate.

The difference in default rates between the two terms is substantial. Loans with a 60-month term exhibit a default rate exceeding 35%, more than double that of the 36-month term, which remains below 17%. This suggests that longer repayment periods may increase borrower risk, possibly due to income volatility or loss of financial discipline over time.

Interest rates follow a similar pattern. The average interest rate for 60-month loans is approximately 16%, notably higher than the 11.7% rate for 36-month loans. This reflects lenders' compensation for the elevated risk and longer exposure.

In short, loan term length plays a critical role in determining both borrower risk and pricing, with longer terms being significantly riskier and costlier.

OLS Regression

The Ordinary Least Squares (OLS) regression model was employed as an initial benchmark to examine how various loan and borrower attributes predict default likelihood. Although not designed for binary classification tasks, OLS regression provides insights into the directional influence and significance of predictors on default status.

The model achieved an R-squared value of 10.3%, indicating it explains a relatively small portion of the variance in default outcomes. Nonetheless, several predictors demonstrated statistical significance. For example, loan amount, installment, annual income, DTI (debt-to-income ratio), and FICO scores all had strong effects. Specifically, higher DTI, longer loan terms, and lower credit scores were positively associated with default, while higher income and stronger credit history were negatively associated.

Categorical features like loan grade and subgrade also yielded clear trends. Borrowers in lower grades (e.g., Grade F and G) showed significantly higher coefficients, reinforcing the creditworthiness gradient embedded in these groupings. The term of the loan (60 months) had a strong positive coefficient of 0.1635, highlighting the elevated risk associated with longer repayment horizons.

Despite these insights, the model's classification performance was limited. Using a threshold of 0.5 to classify defaults, the confusion matrix revealed that while the model correctly identified 475,476 non-defaults and 9,312 defaults, it also misclassified 119,896 defaults as non-defaults. This imbalance led to a modest overall accuracy of 79.24%, largely driven by its performance on the dominant non-default class.

In conclusion, while OLS helped identify influential predictors and offered a broad understanding of variable relationships, its inability to accurately classify defaults—particularly due to the binary nature of the outcome—highlights the need for more specialized classification models, such as logistic regression.

Logistic Regression

To better model the binary nature of the default outcome, a logistic regression was implemented. This model estimates the probability that a loan defaults, offering interpretable coefficients in the form of log-odds, which can be directly related to changes in explanatory variables.

Despite some convergence warnings due to high-dimensional categorical variables and potential multicollinearity, the logistic model still delivered coherent results for most variables. The pseudo R-squared was 10.24%, comparable to the OLS model, suggesting similar explanatory power in terms of variance explained.

As expected, variables such as debt-to-income ratio (DTI), inquiries in the last six months, and loan term positively influenced the likelihood of default. Conversely, annual income and FICO score were strong negative predictors. The coefficient for the 60-month loan term (0.8130) indicates a much higher probability of default compared to shorter-term loans, aligning with the trends observed in descriptive visualizations.

Several issues arose in the model's coefficient estimates, particularly with higher-order categorical levels in loan grade, subgrade, and purpose. These variables exhibited unstable or extreme values (very high standard errors, and uninterpretable magnitudes). This instability likely stems from perfect or quasi-complete separation and multicollinearity due to over-specification of dummy variables.

Nonetheless, the model’s classification performance marginally improved over OLS. With a cutoff of 0.5, the confusion matrix revealed 472,518 true negatives, 12,945 true positives, and 116,263 false negatives, resulting in an accuracy of 79.35%. While this gain is modest, it shows logistic regression's suitability for binary classification compared to OLS.

Overall, logistic regression provided a more appropriate modeling framework for predicting defaults. However, care must be taken in handling high-cardinality categorical variables, potentially requiring regularization or dimensionality reduction in future refinements.

Ridge Regression

Ridge regression was employed to address potential multicollinearity identified in the OLS and logistic regression models. By applying an L2 penalty, Ridge shrinks coefficient estimates to reduce variance without completely eliminating any variables. This regularization technique is

particularly suitable for datasets with a large number of correlated predictors, as is the case in this loan dataset.

The Ridge model was tested across multiple values of the regularization parameter α (alpha), ranging from 0.001 to 10,000. For smaller values of alpha (0.001 to 100), the model's performance remained virtually unchanged. Each of these configurations yielded an accuracy of approximately 79.24%, and identical or near-identical confusion matrices, indicating that moderate regularization had little impact on prediction performance.

As alpha increased beyond 100, a slight drop in accuracy was observed. With alpha = 1000, the accuracy declined marginally to 79.23%, and further decreased to 79.21% at alpha = 10,000. These results suggest that while heavy penalization reduced model complexity, it also led to underfitting, as seen in the increased number of false negatives (loan defaults incorrectly predicted as non-defaults).

Despite the minimal performance improvement over the base OLS or logistic models, Ridge regression contributes value by offering greater model stability and reducing overfitting risk. The consistency in classification metrics across a wide range of alpha values also indicates the robustness of the model.

In summary, Ridge regression offers a balanced approach for handling multicollinearity without significantly compromising predictive accuracy. However, in this context, its performance gains remain marginal, reinforcing the need to complement regularization with other techniques for meaningful improvement.

Lasso Regression

Lasso regression, or L1 regularized logistic regression, was implemented to enhance model interpretability by shrinking irrelevant coefficients to zero. Unlike Ridge regression, which only shrinks coefficients, Lasso can perform feature selection, making it particularly useful for high-dimensional datasets with many potentially redundant variables.

The model was evaluated across five values of the inverse regularization strength C, ranging from 0.001 to 10. As C increases, the regularization effect weakens, allowing more coefficients to enter the model.

At C = 0.001, Lasso produced an accuracy of 79.21%, with relatively high false negatives. As C increased to 0.01 and 0.1, performance improved, with accuracy reaching 79.35%, and the number of correctly identified defaults steadily increasing. This performance plateaued from C = 0.1 through C = 10, suggesting that model performance saturated once sufficient predictors were retained.

Notably, Lasso achieved slightly better precision in identifying defaults compared to Ridge and OLS. At C = 10, the model accurately identified 12,942 defaults, compared to 9,312 in Ridge and 9,312 in OLS, with no sacrifice in overall accuracy. This indicates that Lasso may offer superior performance in detecting minority class outcomes without degrading overall classification.

However, Lasso's computational cost was significantly higher than other models. At larger values of C, training times became prohibitively long, limiting exploration beyond C = 10.

In summary, Lasso regression provided competitive classification performance while offering the added advantage of feature selection. Its ability to retain only the most relevant variables and improve default identification, even with slightly lower overall accuracy at lower C values, makes it a valuable tool when model simplicity and interpretability are key priorities.

5. Be Creative

Beyond the standard regression models implemented in this study, there exist several advanced machine learning algorithms that could potentially yield superior predictive accuracy for loan default classification.

One promising candidate is the Random Forest Classifier, an ensemble method that aggregates predictions from multiple decision trees constructed on random subsets of the data. This approach is particularly effective in handling non-linear relationships and high-cardinality categorical features, such as loan grade, subgrade, and purpose. Moreover, Random Forests are inherently robust to outliers and multicollinearity, and they can accommodate missing data without the need for imputation. These properties make the algorithm well-suited to the structure and scale of the peer-to-peer lending dataset.

Another powerful alternative is XGBoost (Extreme Gradient Boosting). As a sequential ensemble technique, XGBoost builds trees iteratively, with each new tree correcting the residuals

of the previous one. This method has been widely adopted for its high accuracy, computational efficiency, and ability to manage large, structured datasets. With careful hyperparameter tuning—such as adjusting learning rates, tree depth, and regularization penalties—XGBoost could uncover complex patterns in borrower behavior and significantly enhance classification performance.

In this study, we implemented both the Random Forest and HistGradientBoosting models as extensions to our regression-based approach. Among all models tested, HistGradientBoosting achieved the highest overall accuracy at **79.58%**, slightly outperforming both Lasso (79.35%) and Random Forest (79.29%).

In practice, both Random Forest and XGBoost have demonstrated superior accuracy compared to linear models like logistic regression, especially in contexts involving large-scale and high-dimensional data. Given adequate computational resources, implementing and optimizing these models would likely surpass the best performance observed in this study (~79.35% from the Lasso model), offering both stronger classification and better identification of high-risk borrowers.

Furthermore, the predictive outputs of these models can be applied to simulate a practical investment strategy. By leveraging the predicted default probabilities generated by models such as Lasso, it is possible to construct a targeted portfolio with optimized risk-return profiles. The strategy involves filtering loans with a predicted probability of default below a defined threshold (e.g., 20%), thereby reducing exposure to high-risk borrowers. Among the filtered set, loans with the highest interest rates are prioritized to maximize expected yield.

The expected return of each loan can be calculated using the formula:

Expected Return = (1 - Predicted Default Probability) × Interest Rate

By summing across selected loans, a portfolio-level return can be simulated and compared against benchmarks such as investing in all loans or selecting purely based on interest rates. This simulation approach demonstrates how predictive modeling can inform data-driven investment decisions, striking a balance between profitability and risk in peer-to-peer lending.

6. Conclusion

This study demonstrates the viability and importance of predictive modeling in assessing default risk within peer-to-peer lending platforms. By applying multiple regression techniques—including OLS, Logistic, Ridge, and Lasso Regression—to a large, high-dimensional loan dataset, we identified key borrower characteristics that significantly influence loan repayment outcomes. Across all models, income level, debt-to-income ratio, FICO score, loan term, and loan purpose emerged as critical predictors of default. While logistic regression provided a more appropriate framework for binary classification than OLS, Lasso regression achieved the highest predictive accuracy (79.35%) and enhanced model interpretability through effective feature selection. These findings underscore the value of regularization in managing complex financial data and highlight the potential for data-driven tools to improve risk evaluation and lending decisions. Our results offer practical insights for P2P investors and platform designers aiming to optimize portfolio performance while promoting responsible and inclusive credit allocation.

7. References

Chang, A. H., Yang, L. K., Tsaih, R. H., & Lin, S. K. (2022). Machine learning and artificial neural networks to construct P2P lending credit-scoring model: A case using Lending Club data. *Quantitative Finance and Economics, 6*(2), 303–325. https://doi.org/10.3934/QFE.2022013

Munkhdalai, L., Munkhdalai, T., Namsrai, O., Lee, J. Y., & Ryu, K. H. (2019). An empirical comparison of machine-learning methods on bank client credit assessments. *Sustainability, 11*(3), 699. https://doi.org/10.3390/su11030699

Serrano-Cinca, C., Gutiérrez-Nieto, B., & López-Palacios, L. (2015). Determinants of default in P2P lending. *PLOS ONE, 10*(10), e0139427. https://doi.org/10.1371/journal.pone.0139427